



GENERAL APTITUDE

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Number System

Q. A 3-digit number $4a5$ is added to another 3-digit number 675 to give a 4-digit number $11b0$, which is divisible by 3. Then $a+b$ is:

A. 16

B. 17

C. 15

D. 14

Ans : A



Number System

Q. In a two digit prime number, if 18 is added we get another prime number with reversed digits. How many such numbers are possible?

- A. 1 B. 2 C. 3 D. None of these

Ans : B



PRIME NUMBERS

1	<u>11</u>	21	<u>31</u>	<u>41</u>	51	<u>61</u>	<u>71</u>	81	91	5
<u>2</u>	12	22	32	42	52	62	72	82	92	1
<u>3</u>	<u>13</u>	<u>23</u>	33	<u>43</u>	<u>53</u>	63	<u>73</u>	<u>83</u>	93	7
4	14	24	34	44	54	64	74	84	94	0
<u>5</u>	15	25	35	45	55	65	75	85	95	1
6	16	26	36	46	56	66	76	86	96	0
<u>7</u>	<u>17</u>	27	<u>37</u>	<u>47</u>	57	<u>67</u>	77	87	<u>97</u>	6
8	18	28	38	48	58	68	78	88	98	0
9	<u>19</u>	<u>29</u>	39	49	<u>59</u>	69	<u>79</u>	<u>89</u>	99	5
10	20	30	40	50	60	70	80	90	100	0



Averages

Q. The average weight of 16 boys in a class is 50.25 kg and that of the remaining 8 boys is 45.15 kg. Find the average weights of all the boys in the class.

A. 47.55 kg

B. 48 kg

C. 48.55 kg

D. 49.25 kg

Ans: C



Averages

- **Simple Average :**

- An average of a set of values is the sum of values divided by the total number of values.
- Average of 'n' values = (Sum of the 'n' values)/n
- This is also called as Arithmetic Mean.
- Average (A) = Sum (S)/ Number(n)
- $S = A \times n$

- **Weighted Average :**

- When all values whose average we want to find do not have uniform occurrences we calculate the weighted average.
- If values $y_1, y_2, y_3 \dots$ occur $w_1, w_2, w_3 \dots$ times then
- Weighted Avg =
$$\frac{(w_1y_1 + w_2y_2 + w_3y_3 + \dots)}{(w_1 + w_2 + w_3 \dots)}$$



Averages & Ages

Q. The average age of husband, wife and their child 3 years ago was 27 years and that of wife and the child 5 years ago was 20 years. The present age of the husband is:

A. 35 years

B. 40 years

C. 50 years

D. 60 years

Ans: B



Ages

Q. The sum of the present ages of a father and his son is 60 years. Six years ago father's age was five times the age of the son After six years son's age will be-

A. 12 years

B. 14 years

C. 18 years

D. 20 years

Ans: D



Profit & Loss

Q. A shopkeeper has 50 kgs of rice. He sells a part of it at 20% profit and the rest at 40% profit. If he gains 25% on the whole, find the quantity of each part.

- A. 12.5 kgs and 37.5 kgs
- B. 37.5 kgs and 12.5 kgs
- C. 23.5 kgs and 21.5 kgs
- D. 21.5 kgs and 23.5 kgs

Ans: B



Mixtures & Allegations

Q. What quantity of sugar costing Rs 6.10 per kg must be mixed with 126 kg of sugar priced at Rs 2.85 per kg so that 20% may be gained by selling mix at Rs 4.80/kg ?

A. 65 kg

B. 69 kg

C. 66 kg

D. 64 kg

Ans: B



Mixtures & Alligation

Type 1 oranges at Rs.60 per kg and Type 2 oranges at Rs.120 per kg and when mixed cost is Rs.75 per kg. Find the ratio in which Type 1 and Type 2 oranges are mixed.

Soln:

Type 1
60

Type 2
120

75

$$x = d - m$$

$$y = m - c$$

$$\frac{x}{y} = \frac{d - m}{m - c} = \frac{120 - 75}{75 - 60} = \frac{45}{15} = \frac{3}{1} = 3:1$$

CP of cheaper
ingredient (c)

CP of costlier
ingredient (d)

Mean Price (m)

CP of costlier ingredient
- Mean Price

Mean Price - CP of
cheaper ingredient

$$\frac{\text{Quantity of cheaper ingredient}}{\text{Quantity of costlier ingredient}} = \frac{d - m}{m - c}$$



Mixtures & Allegations

Q. A container has 60 litres of milk. From this container 6 litres of milk was taken out and replaced by water. The process was further repeated twice. The volume of milk in the container after that is

- A. 38.52 lit B. 43.74 lit C. 42.68 lit D. 39.85 lit
- E. None of these

Ans: B



Mixtures & Alligation

- **Final concentration = Initial $(1 - \frac{R}{\text{Initial}})^n$**
- where,
- Final concentration is the amount of concentration remaining after the process
- n is the number of times the process is done and
- R is the replaced quantity.
- Initial is the initial concentration



Mixtures & Alligation

Q. A container contains 40 litres of milk. From this container 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?

- A. 26.34 litres B. 27.36 litres C. 28 litres D. 29.16 litres

Ans: D

- The volume of milk remaining after the three processes is,

- $V = N \left(1 - \frac{R}{N}\right)^n$

where,

N is the original amount of milk,
n is the number of processes and
R is the replaced quantity.(taken out quantity)

- $= 60 \left(1 - \frac{6}{40}\right)^3$

- $= 60 \left(1 - \frac{1}{10}\right)^3$

$$= 60(0.729)$$

$$= 43.74 \text{ lit}$$



Profit & Loss

Q. If the cost price of 6 pencils is equal to the selling price of 5 pencils, then the gain percent is

A. 10%

B. 20%

C. 15%

D. 25%

Ans: B



Interest

Q. The compound interest on a certain sum for 2 years at 10% per annum is Rs. 525. The simple interest on the same sum for double the time at half the rate percent per annum is:

A. Rs. 400

B. Rs. 500

C. Rs. 600

D. Rs. 800

Ans: B



Partnership

Q. In a partnership A puts Rs. 2000 for entire year, B puts Rs. 3000 initially & increases it to 4000 at the end of 4 months, c puts in Rs. 4,000 initially but withdraws Rs 1000 at the end of 9 months. What is C's share of annual profit of Rs. 8475?

A. Rs. 1800

B. Rs. 3300

C. Rs. 3375

D. Rs. 3350

Ans : C



Ratio & Proportion

Dividing a given number in the given Ratio :

Let A be the given number. Let the given ratio be a:b:c

This means A is divided into three parts such that

$$\text{First Part} = A \times a/(a+b+c)$$

$$\text{Second Part} = A \times b/(a+b+c)$$

$$\text{Third Part} = A \times c/(a+b+c)$$

$$\text{And First Part} + \text{Second Part} + \text{Third Part} = A$$

$$\text{Any Part} = \text{Total Amount} \times (\text{Its related ratio term} / \text{Sum of Ratio Terms})$$



Interest

Q. A began a business with Rs. 85000. He was joined afterwards by B with Rs. 42500. After how many months did B join if the profit at the year is divided in the ratio of 3:1?

A. 4 months

B. 5 months

C. 6 months

D. 8 months

Ans: A

using formula,

$$\frac{C_1 T_1}{C_2 T_2} = \frac{P_1}{P_2}$$

In this type , the time period is 12 months i.e. one year



Time & Work

Q. A & B can complete a work in 30 & 40 days respectively. But with the help of C they completed the work in 16 days & they are paid Rs 4500. Find the share of C in the pay

A. Rs. 300

B. Rs.1800

C. Rs. 2000

D. Rs. 600

Ans: A



Pipes & Cisterns

Q. An inlet pipe can fill a tank in 24 hours but because of a leak, it takes 16 hours more to fill the tank. In how many hours can the leak empty the tank?

A. 40 hrs

B. 48 hrs

C. 60 hrs

D. 120 hrs

Ans: C

Time needed to do the work = $\frac{XY}{Y-X}$ where,

X = number of hrs to fill tank ,

Y = number of hrs to fill tank with leakage



Time & Distance

Q. A person travels 285 km in 6 hours in two stages. In the first part of the journey, he travels by bus at the speed of 40 km per hour. In the second part of the journey, he travels by train at the speed of 55 km per hour. How much distance did he travel by train?

A. 145 km

B. 165 km

C. 185 km

D. 205 km

Ans: B



Time & Distance

Q. Shekhar reaches office 20 min late while walking at a speed of 20km/hr. Next day he increases his speed by 5km/hr but is still late by 10 min. what is the distance of his office from his home?

A. 16.67km

B. 30 km

C. 12 km

D. 20 km

Ans: A



Time & Distance

Q. Ram & Shyam stay 30 km apart. They decide to meet between their houses. Ram starts at 3 pm & cycles at 4 km/hr & Shyam starts at 5 pm & cycles at 6 km/hr. At what time do they meet?

A. 6:15 pm

B. 7:10 pm

C. 7:12 pm

D. 7:22 pm

Ans : C



Calendar

Q. On what dates of August 2020 will Monday fall :

A. 3,10,17,24,31

B. 5,12,19,26

C. 1,8,15,22,29

D. None of these

Ans: A



Clock

Q. The minute hand of a clock is 12cm long. Then the area of the face of the clock described by the minute hand in 35 minutes is:

A. 210 cm²

B. 264 cm²

C. 144 cm²

D. 200 cm²

Ans : B

$(\theta/360)\pi r^2$ is the area of the sector of a circle of radius r and central angle θ .



Races

Q. A, B & C start together running along a circular track of 500 m at 8 km/hr, 5 km/hr & 3 km/hr respectively. After how much time will all three meet again at the starting point for the first time?

A. 20 min

B. 24 min

C. 30 min

D. 36 min

Ans: C

Step1 – find the time taken by each member /player to complete 1 round

Step2 – Calculate LCM(Lap time)



Circular Motion

- Use of both relative speed & LCM
- Let S_a, S_b = speeds of two persons.

S_r = Their relative speed

Distance traveled in 1 round = circumference

Case A : Both running in Same direction

Both meet again first time when $\rightarrow$ **Time = $\text{dist}/S_r = \text{Circumference}/S_a - S_b$**

Case B : Both running in opposite directions(**$\text{DistA} + \text{DistB} = \text{Circumference}$**)

Both meet first time when $\rightarrow$ **Time = $\text{Circumference}/S_a + S_b$**

Case C : Both running in same/opposite directions

Both meet again at starting point at LCM of their Lap times.



